

Fire Alarm Using a Thermistor

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This project can effectively be used as a temperature-sensing fire alarm. Here, an NTC thermistor is used as the temperature sensor. Resistance of the NTC thermistor decreases with an increase in temperature.

Circuit and working

Circuit diagram of the fire alarm is shown in Fig. 1. It is built around NTC thermistor (NTC1), transistor

BC547 (T1), popular NE555 timers (IC1 and IC2), speaker and a few other components.

In this circuit, the two NE555 timers are wired as astable multivibrators. IC1 is wired as a low-frequency generator and IC2 as a high-frequency generator.



PARTS LIST

Semiconductors:

- IC1, IC2 - NE555 timer
- T1 - BC547 npn transistor
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 22-kilo-ohm
- R2 - 4.7-kilo-ohm
- R3, R6 - 1-kilo-ohm
- R4, R7 - 56-kilo-ohm
- R5 - 680-ohm
- R8 - 10-ohm
- VR1 - 100-kilo-ohm potentiometer

Capacitors:

- C1, C6 - 100 μ F, 25V electrolytic
- C2 - 2.2 μ F, 25V electrolytic
- C3, C5 - 0.01 μ F ceramic disk
- C4 - 0.047 μ F ceramic disk

Miscellaneous:

- CON1 - 2-pin connector terminal
- S1 - On/off switch
- NTC1 - 10-kilo-ohm NTC thermistor
- LS1 - 8-ohm, 0.5W speaker

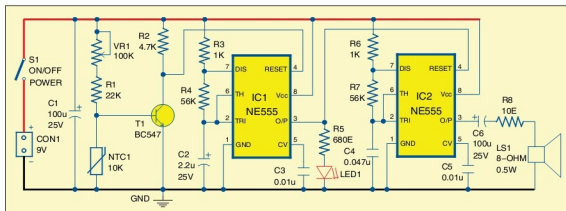


Fig. 1: Circuit diagram of the fire alarm

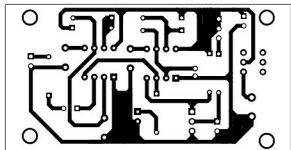


Fig. 2: Actual-size PCB pattern of the circuit for the fire alarm

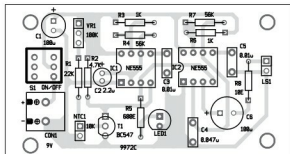


Fig. 3: Component layout of the PCB shown in Fig. 2

At room temperature, transistor T1 conducts and keeps reset pin 4 of IC1 at ground level. As a result, both timer ICs are disabled. But when temperature of the sensor goes above 70°C (depending on the thermistor constant [K]), transistor T1 stops conducting. Both NE555 timers oscillate and a beeping sound is heard from the speaker.

Potmeter VR1 is used to set the cut-off/saturation condition of transistor T1, which is related to the resistance of NTC1 at different tem-

peratures. At room temperature, voltage at pin 4 of IC1 remains low. With heating of NTC1, voltage at pin 4 of IC1 becomes high. This enables both the timer ICs to oscillate and produce sound through the

speaker. Also, LED1 starts flashing.

The circuit works on 9V regulated power supply. For a louder sound, you may add a speaker driver circuit with matched impedance.

Construction and testing

An actual-size, single-side PCB for the fire alarm circuit is shown in Fig. 2 and its component layout in Fig. 3. Enclose the PCB in a small box in such a way that the thermistor can sense the temperature in case of fire. ●



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